Muhammad Aulia Rahman Santosa

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Profile Summary

Recent Mechanical Engineering graduate from Brawijaya University (GPA 3.28/4.00) with a thesis on tensile strength optimization of dissimilar friction-welded joints (Al 6061 to ST60 steel) using Response Surface Methodology in Minitab, and a foundation in equipment maintenance, root cause analysis (RCA), and HSE practices. Contributed to a maintenance improvement project at APRIL Group using Fishbone, 5 Why, and PDCA methods that reduced overhaul downtime by 50%. Supervised multi-machine laboratory operations and led large-scale organizational projects. Eager to apply mechanical engineering fundamental, equipment maintenance, and leadership skills to drive operational efficiency in engineering environment.

Education

Brawijaya University Malang, Indonesia
Bachelor of Engineering in Mechanical Engineering 2022 - 2026
GPA: 3.28/4.00
Thesis: Optimization of Tensile Strength in Dissimilar Friction Welded Joints of Aluminium 6061 and St60 Steel Based on Single-Sided Chamfer, Friction Time, and Initial Friction Pressure.
Modeled and optimized welding parameters using Minitab Response Surface Methodology Box-Behnken Design, achieving optimal ultimate tensile strength of 235.38 MPa (20° chamfer, 8s friction time, 4.2 kN pressure) with model accuracy (R^2 = 98.01%).

Professional Experience

Riau Andalan Paperboard International (APRIL Group) Riau, Indonesia
Mechanical Maintenance Intern July 2025 – August 2025

- Participated in the Board Machine 1 (BM1) roll cleaning facility improvement project, analyzing root causes of maintenance bottlenecks using Fishbone Diagram and 5 Why Analysis and documenting the findings.
- Documented the results of the Kadant Unitrak sliding rail adapter system implementation, which reduced overhaul downtime by 50% (6 to 3 hours) and technician requirement by 50% (10 to 5 per overhaul).
- Drafted a proposed SOP using the PDCA cycle and assisted in verifying PTW and JSA documentation checklists prior to maintenance work, supporting safety compliance throughout the improvement project.

Manufacturing Process Laboratory I Malang, Indonesia
Laboratory Assistant and Infrastructure Manager June 2024 – July 2026

- Mentored and guided 130+ engineering students across 4 semesters, conducting quality assurance (QA) audits and technical evaluations on practicum reports to enhance analytical accuracy, report quality, and overall understanding of manufacturing processes.
- Diagnosed and resolved a mechanical malfunction across 7+ conventional machines such as milling machine, lathe machine, and welding machine restoring full operation and preventing disruption to scheduled practicum sessions.
- Coordinated and supervised 4 practicum modules per semester for 80+ students, providing safety compliance and hands-on guidance to achieve effective learning outcomes and safe laboratory operations.

Leadership & Organizational Experience

BEM Teknik Universitas Brawijaya Malang, Indonesia
Executive Board Member (Collective Presidency) - Head of External Affairs Dept. September 2025 – September 2026

- Co-led a collectively governed faculty-level executive board of 10 members (no single president), setting strategy and institutional alignment across 8 engineering departments.
- Led the External Affairs Department: managed 28 staff and 6 strategic programs, delivering 28 media partnerships and 3 corporate MoUs (OYI Group, Samperkap.mlg, Selfroom Group) meeting the department's KPI target.
- Managed stakeholder relations and alumni outreach: tracked 1,300+ alumni across 8 departments, coordinated 2+ university exchange sessions (TEKSAS), and published 2 alumni feature podcasts.

Kemah Kerja Mahasiswa XLV FT-UB Malang, Indonesia
Chief Executive November 2024 – February 2025

- Directed the planning and on-site program execution of a faculty-wide student orientation and community service program involving 1,180+ new students and 390+ committee members across 7 field divisions.
- Led coordination with university authorities, village officials, vendors, and local communities in stakeholder coordination for technical and infrastructure-based programs, successfully delivering 7 community development key projects including drainage normalization, paving installation, public street lighting installation, maintenance of tourist attraction, education programs (5 schools), craftmanship, and tree planting.
- Managed site safety, resource allocation, and quality control, monitoring team performance and field progress to ensure on-time project delivery within technical specifications.

Student Organizations and Event Committees | 2023 – 2025 Malang, Indonesia

- Staff of External Affairs, BEM Teknik UB (2023 – 2025): led publication, documentation, and visual design for faculty visit programs (PT PAL, Universitas Diponegoro) and served as PIC of an internal campus visit program. Also supported faculty events and served as Master of Ceremony for organizational activities.
- Staff of External Affairs, HMM FT-UB (Mar 2024 – Feb 2025)
- Department committees: M-Games, Event Division (closing team, 2023); Probinmaba Mesin, participant monitoring (sweeper) team (2023) and participant evaluator (2024).
- Faculty committees: PIC of public street lighting installation (Kemah Kerja XLIV, 2024); Mentoring and Materials Team (Pelatihan Kader Teknik, 2024); Security Team (KRIDA, 2023) and Evaluation Team (KRIDA, 2024); Equipment and Security Team (Kongres Mahasiswa Teknik, 2023).

Project Experience

GreenSense: Smart Greenhouse Irrigation System IoT with CNC Plotter and Machine Learning

Developed a precision greenhouse irrigation system using IoT, integrated with CNC plotter movement and machine learning for optimized, automated water distribution, achieving up to 50% water savings. Led the mechanical fabrication, structural assembly, and precision alignment of the CNC plotter movement mechanism.

Core Competencies

- Manufacturing Process and Quality Assurance
- Project Monitoring & Coordination (On-site)
- Equipment Maintenance & Continuous Improvement
- Root Cause Analysis & Problem Solving

Technical & Digital Tools

- Microsoft Word, Excel, PowerPoint
- AutoCAD (Basic)
- SolidWorks (Basic)
- ImageJ
- Minitab (DOE and statistical analysis)
- Autodesk Inventor (Working knowledge)

Certification & Training

- **Basic Industrial Welding** | SPPVP Lampung (Kemnaker RI) - August 2025
- **TOEFL ITP Prediction – 567** | June 2026
- **EF SET English Certificate (Reading and Listening)** – 72/100, C2 Proficient | August 2025